BIRZEIT UNIVERSITY

Faculty of Engineering and Technology Computer Science Department

COMP433

Software Engineering

Final Project Report

Group Name: CyberSquad

Bakery Shop Online

Prepared By:

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Instructor's Name: Dr. Adel Taweel

Chapter One: Project Planning and Management.

1.1. Names of Editors/writers of the Report:

- Dunia Jaser
- Hala Ziq

1.2. Business Title:

- **Bakery Shop Online**

1.3. Group Name and Number:

G4: "CyberSquad" is chosen to be the name of our group

1.4. Group Members:

1. Dunia Jaser – 1201345.
2. Hala Ziq – 1191637.
3. Ansam Hababeh – 1201127.
4. Aya Abdalmuti – 1212074.
5. Rama Shaheen – 1211153.

1.5. Members Roles:

1. **Project Manager**: Hala Ziq
2. **QA or Software Tester:** Dunia Jaser
3. **Secretary:** Ansam Hababeh
4. **Technical Architect:** Rama Shaheen
5. **Programmer:** Aya Abdalmuti

1.6. Project Management Strategy:

Meetings:

Our team met bi-weekly on Wednesdays and Saturdays via zoom to discuss project progress, issues, and allocate tasks, and sometimes we created face-face meeting on the university.

Decisions:

The project manager made final decisions based on group discussions (Voting), where the secretary assisted in organizing meetings, and following up on assigned tasks.

Software Process models:

The team has adopted the Agile software development model. This approach has allowed us to remain flexible and responsive to changes throughout the development process, ensuring continuous improvement and iteration based on feedback and the evolving project needs.

1.7. Project manager report:

At the beginning, we divided the work among the group members equally and fairly. Each member was responsible for their individual tasks, while we collaborated on group tasks to ensure consistency and quality in our project.

Hala Ziq (Project Manager): I was responsible for project planning, scheduling, also ensured all team members stayed on track and met their deadlines, support and help the busy member to submit the overall phase on time. Where I led and wrote most of the **group tasks** like **User and System Requirements**, **Effort, Time and cost Estimation**, **Actors' roles and their semantic descriptions**, **over all Activity Modelling**, **over all use case**, **Component modelling**, and **System and architectural Design**, I was responsible to review and discuss all of the tasks even it was group or individual tasks, for the individual tasks I write the **Admin Edits Product Information scenario**, I wrote the detailed use-case specification, activity modelling, and sequence diagram for this scenario taking the notes which provided by DR. Adel on the point.

- **Team Members and Contributions:**

1. **Dunia Jaser (QA/Software Tester):** Dunia played an important role in ensuring the quality and robustness of our system. She led the tasks related to the Product Browsing sequence diagram and the analysis and design of Scenario 2. Dunia also wrote and reviewed the Product Browsing Use-Case Specification. Her contributions extended to discussing and reviewing actors and use case diagrams, providing feedback for the System Class Diagram, participating in group discussions to refine System Requirements, and ensuring the accuracy and completeness of the project.

2. **Aya Abdalmuti (Programmer):** Aya led and write the Customer Gives His Feedback scenario. She wrote the detailed use-case specification, activity modelling, and sequence diagram for this scenario. Aya also participated in discussions on effort estimation, requirement analysis, architectural design, and deployment modelling. She reviewed business descriptions and power point presentations, contributing significantly to the project's documentation and technical integrity.

3. **Rama Shaheen (Technical Architect):** Rama led the scenario of the payment processing, she created the detailed use-case specification, activity model, and sequence diagram for the payment process. Rama also contributed to system class modelling and provided valuable insights during team discussions, helping to shape the project's direction and technical architecture.

4. **Ansam Hababeh (Secretary):** Ansam contributed to various phases of the project. She discussed with the other members the user and system requirements in phase 2. She was responsible for effort, time, and estimation calculations. Ansam wrote and reviewed use-case specifications for sign up, login processes, and other scenarios. She edited use case modelling diagrams, drew sequence diagram, and system and deployment model, and designed the project presentation.

- Managing this project presented several challenges like balancing tight deadlines with the phases requirements which required effective time management and priority on the work, the different free times for the members sometimes led to delays in the task submission and the feedback for the reviewed tasks, modification and edit on the diagrams takes time because we wasn't familiar with drow.io tool.
- **Evaluation of Project Success:**

In my opinion, our project has been successful. We achieved our primary objectives by delivering all phases at the time and doing good at the project presentation, the management strategy allowed us to improve continuously based on DR. Adel feedback, which was critical to our success. The collaborative efforts of our team members in overcoming challenges and meeting deadlines played a significant role in our achievements.

1.8. Group members report:

Dunia Jaser

I believe our project, the online bakery shop system, has been successful. The project has met all our primary objectives, providing a user-friendly platform for customers to browse and purchase bakery items seamlessly via a mobile app. The system's robust backend, coupled with a responsive frontend, ensures a smooth user experience on mobile devices. Our adoption of the Agile methodology allowed us to iteratively improve the system based on continuous feedback, which was crucial to our success. Additionally, the collaborative effort of our team in tackling challenges and ensuring the timely completion of tasks contributed significantly to the project's overall success. I would like to express my gratitude to Dr. Adel for his mentorship and for providing us with the valuable feedback that was essential for our success.

Contribution to the Project:

- **Led Tasks:**
  - Drawing and designing the **Product Browsing** sequence diagram.
  - Leading the analysis and design of **Scenario 2: Product Browsing**.
  - Writing and reviewing the **Product Browsing Use-Case Specification**.
- **Contributed Tasks:**
  - Discussing and reviewing the **Actors and Use Case Diagram**.
  - Providing inputs and feedback for the **System Class Diagram**.
  - Participating in group discussions to refine the **System Requirements**.
  - Reviewing and providing suggestions for the **System and Deployment Modelling**.
  - Checking and ensuring the accuracy and completeness of the entire project.

Aya Abdalmuti

In my view, this project was a valuable experience that allowed me to actively participate in collaborative work. Working closely with team members, I had the opportunity to contribute to various aspects of the project. It is really useful in several areas for the future, whether for the labour market or the graduation project. This course has not only enhanced my technical skills but also instilled confidence for tackling future challenges in project work. I extend my sincere gratitude to Dr. Adel, who has greatly benefited us by providing advice and feedback.

- Write SCENARIO for my use case which was Customer Gives His Feedback.
- Detailed Use-Case Specification for Customer Gives His Feedback.
- Activity Modelling for Customer Gives His Feedback.
- SEQUENCE Diagram for Customer Gives His Feedback.
- Discussion in Effort Estimation.
- Review Business description.
- Discussion in Requirement Analysis.
- Discussion in architectural design, class diagram, deployment modelling.
- Detailed Class Diagram Contributed by my.
- Review power point.

Rama Shaheen

My point of view, as a partner in this group, I believe we worked well together as a team to accomplish our project's objectives, this project was successful we worked together as a team, it gave me good knowledge to use in other projects in the future, my primary contribution was leading the payment processing scenario, making sure it was reliable and efficient.

- Write a scenario for my use case (payment process method)
- Detailed use case specification for payment process method.
- Activity model for payment process method.
- System class modeling
- Detailed class diagram
- Sequence diagram for payment process method
- Actively participated in team discussions, providing insights and suggestions that helped shape our project's direction.

Ansam Hababeh

To me, the success of our software project is evident from the feedback we received and the skills we developed during the process. We effectively addressed all the project requirements and delivered a functional, well-designed application. The collaboration and problem-solving skills we gained are invaluable, and the positive responses from our instructor indicate that our project was successful and well-executed. I am grateful to Dr. Adel for his mentorship and for providing us with the feedbacks needed to succeed.

- Write some points in phase1 for agriculture shop online.
- Write some of user requirement and system requirement in phase 2.
- Calculate Effort, time and estimation for phase 2.
- Write Scenario of user Account management and reviewed my team scenarios.
- Reviewed Actors and use cases for bakery system
- Write use-case specification of sign up and login process and reviewed dunia and hala.
- Edit use case modelling diagram.
- Draw use case activity modelling for sign up process and reviewed add product modelling.
- Draw sequence diagram for sign up.
- Write System Design Goals.
- Draw System and deployment modelling.
- Design presentation.

Chapter Two: Requirement Elicitation, Analysis and Modelling.

2.1. Business Description:

The online bakery shop makes it easy for customers to order a variety of bakery and bread products from home. They offer cakes, pastries, muffins, artisanal bread, and special items for different tastes and diets. Everything is delivered straight to the customer's door, ensuring convenience and freshness.

The bakery can handle a large number of orders, ranging from hundreds to thousands each month. They have a wide selection of products, with 50 to 600 different items available at any time. Sales for each product vary, from hundreds to thousands of units per month, depending on popularity. Using the service is simple. Customers can visit the bakery's mobile app to browse and choose products. They add items to a virtual shopping cart and then proceed to checkout, where they can select a delivery time and payment method, including cash on delivery or online payments. Once an order is placed, the bakery prepares the products fresh. The order is then packaged and sent out, with delivery charges based on location and order size. This ensures customers get their bakery items fresh and conveniently delivered to their door, combining ease with quality and satisfaction.

The bakery administration manages everything by overseeing orders, inventory, and deliveries. They keep track of incoming orders and ensure they are fulfilled on time. They monitor stock levels to make sure they have enough ingredients and products available. They also coordinate with the baking team to ensure all items are freshly made and packaged correctly. Additionally, they handle customer service, addressing any issues or concerns that arise. By keeping everything organized and running smoothly, the administration ensures the bakery operates efficiently and customers stay happy.

2.2. User and System Requirements (Led and Written by Dunia and Hala, discussed by all)

UR1.0: The system shall allow users to create, login the account, and manage the user profile.

- **SR1.1:** The system shall provide a registration form to fill the username, email, password, phone number and address, then validated the filled information.
- **SR1.2:** The confirmation for the phone number should be sent to the user in SMS message, after authenticate the phone number the user information will be saved.
- **SR1.3:** The users shall be able to log in by their registered email and password, after authenticated their credentials, if the user is not authenticated the system will require from the user to register.
- **SR1.4:** The system shall allow users to manage the profile by edit their personal information like username, password, email, phone number, and address, then save the changes.

UR2.0: A user shall allow the user to browse and search the bakery products, with each product displaying detailed descriptions, ingredients, and prices.

- **SR2.1:** The system shall display a menu of bakery products available for sale, and mark the products that will be out of stock soon.
- **SR2.2:** Each product on the menu shall include detailed descriptions about the ingredients, prices, count on the stock, and images.
- **SR2.3:** Users shall be able to browse products by categories or by search functionality to find specific products.
- **SR2.4:** The system shall provide filters and sorting options to help users limiting their search results, such as : popularity, new arrivals,

UR3.0: The system shall allow users to customize their orders by selecting product variations such as: (e.g., cake Flavours, bread sizes), choosing delivery times, and confirm orders.

- **SR3.1:** Users shall have the ability to customize their orders, such as selecting product variations, specifying delivery times, and adding special instructions.

- **SR3.2:** The system shall make the order in real-time based on user customization selections.

UR4.0: The system shall provide virtual shopping cart to allow the user able to add or remove products before purchase.

- **SR4.1:** The system shall provide a virtual shopping cart to manage (add, remove, and update) the product selections before confirm the order.
- **SR4.2:** The shopping cart shall display the total price, quantity of each product, and option to confirm the order.

UR5.0: The system shall offer multiple payment methods for users including credit card, and cash on delivery.

- **SR5.1:** The system shall support multiple payment methods, including credit card payments and cash on delivery.
- **SR5.2:** if the user selects the online payment, the system shall securely save the credit card information for future transaction, if the payment process takes more than one minute the system will not confirm the order or take a money from the user (order cancelation) and offer cash on delivery payment.
- **SR5.3:** The system shall check if the users is a regular customer ask them to take a discount by using the loyalty card, if the user choose to use it they will have a discount.

UR6.0: A user shall be able to view their purchase history.

- **SR6.1:** Users shall be able to view their past orders, including order details such as items purchased, quantities, prices, order dates, and delivery details.

UR7.0: The system shall provide a feature for users to submit feedback on products and the service they receive after each order.

- **SR7.1:** Users shall be able to submit feedback and ratings for each product purchased and the service after each order.
- **SR7.2:** The system shall record and display user feedback and ratings for the products to make improvement purposes.

UR8.0: The system shall provide notifications and alerts for the users when add a new products, discounts or changes to the users' orders or delivery details, and for admins when receive new order or products nearly will be out of stock.

- **SR8.1:** The system shall send notifications and alerts to users when new product, discounts added, product will be out of the stock soon, changes on the orders or on the delivery details.
- **SR8.2:** Admins shall receive notifications for new orders and when the products almost out of stock.

UR9.0: The system shall provide administration panel to allow the admins to monitor, and analyze various aspects of the bakery's operations, manage the inventory by restocking and products availability status, make discounts, edit products information, and reporting features for detailed business insights.

- **SR9.1:** Admins shall be able to monitor various of the bakery operations, by managing the products on inventory.
- **SR9.2:** Admins shall be able to apply discounts, add new products, edit product information.
- **SR9.3:** The system shall automatically generate a reports and analysis for business insights.

- **SR9.4:** Admins shall receive notifications for new orders and when the products almost out of stock.

2.3. Scenario Analysis

Scenario 1: User Account Management (Ansam Hababeh) (MODIFIED)

Initial Assumption:

Users must sign up and log in to the system to access and interact with the platform's features. The system must have the necessary infrastructure installed, including integration with an SMS gateway for sending verification codes and verifying phone numbers. This service can be invoked when the system is fully operational, with all dependencies and configurations properly set up, and users are authenticated and authorized to use the platform's features.

Normal:

Typically, users start by accessing the business application and selecting the sign-up button. This action brings up the registration page, where users are required to enter their information, including a username, password, phone number, and address. Once the user submits their details, the system verifies the entered data and sends a verification code via SMS to the provided phone number. The user must then enter the received code on the registration page. Upon successful entry of the correct code, the system verifies the phone number and saves the user information, creating a new account. The user can then log in to the application with their credentials.

Alternative:

Users who have already registered on the platform can access the login page by entering their email and password, instead of going through the registration process. Once the system successfully authenticates their credentials, the user is granted access to their account, allowing them to continue browsing and using the platform's services. If the authentication fails, the system will prompt the user to complete the registration process.

Additionally, there are alternative ways to interact with the platform's functionalities. Users can either search for specific products using the search bar or browse through different categories for a more exploratory shopping experience. For payments, users have the option to pay by cash or by card, providing flexibility in how they complete their transactions.

Error:

Possible errors happen when user sends an incorrect or incomplete registration forms. For example, if user leave an empty field or enters incorrectly formatted email address, the system will display the error on screen like(Invalid email format or Password must be at least 8 characters long).The system allow user to request a new SMS and attempt the verification again if the phone number verification fails, for example, because the code that get from SMS was entered incorrectly.

Other Activities:

In the System the connections are established, security measures are integrated, user data is secured, and Two-factor authentication for login is implemented to enhance security.

System State on Completion:

After successful registration or login, the system gives users access to their profile and interact with system features including manage their profile and edit information. They can view the bakery product and make order. The system store users and is ready to receive another request.

Scenario 2: Product Browsing (Dunia Jaser) (MODIFIED)

Initial Assumption:

The customer has successfully logged into the online bakery store and is now on either the home page or a page about a product, ready to browse bakery items. This browsing service is only available when the network connection and the system's backend are stable, which includes product files and inventory management. This makes sure that users can easily access the most up-to-date information and pictures about products.

Normal:

Whenever a customer visits to the bakery's online store, they access the main navigation page to look through product types like bread, cakes, and pastries. Each category is automatically filled with products that show a lot of information, like prices, high-resolution pictures, complete descriptions, and a list of ingredients. The customer picks out from the list the item they want, like a chocolate cake. Then, they are taken to a thorough page about the product that has more information and, if needed, customization choices like choosing the size or flavour. The customer can add the item to their virtual shopping cart after choosing the changes they want. The system shows that the item has been added and changes the cart's contents to reflect this are involved. After that, the customer can either keep looking for more items or go to the checkout page. The system's browsing features are meant to make buying easier by making it easy to find products and handle the shopping cart. This process shows how the system's browsing features should normally be used.

Alternative:

If a customer finds their desired product temporarily out of stock or a particular variation not available, the system facilitates two clear alternative routes for product selection. First, the system can redirect the customer to a search functionality where they can enter specific criteria or keywords, such as "chocolate desserts" or "birthday cakes," to find similar or alternative products. This search function uses advanced filtering options, including flavour, size, and price range, allowing the customer to adapt their search results more closely to their needs.

Secondly, if the customer prefers browsing rather than searching, they can utilize the “related products” feature which dynamically suggests similar items based on the features of the missing product the customer is looking for. For example, if a chocolate cake is out of stock, the system might suggest a double chocolate cake or a chocolate fudge cake as alternatives, displayed prominently alongside the out-of-stock notification. This way, the customer can continue their shopping experience without the need to manually search for alternatives, enhancing user satisfaction and keeps them interested in the online store.

Error Scenario:

During the product browsing process, an error might occur if the system fails to retrieve product details from the database due to a server or database malfunction. This could happen, for example, when multiple users simultaneously request data for the same product, overloading the server. In this scenario, the customer may experience a delay or receive incorrect product information, which could ruin their buying experience. To address this, the system will display an error message informing the customer of the technical difficulty and suggest trying the action again later. Additionally, the system logs such incidents for the technical team to analyse and rectify, aiming to enhance database handling and prevent similar failures in the future. If the issue persists, the system might also temporarily disable the affected product's page until the error is resolved, ensuring that no incorrect transactions can be made. This proactive approach helps maintain the reliability of the shopping experience and reduces potential frustration for the customer.

Other Activities:

In the online bakery store, while customers engage in the primary shopping process, they can also partake in supplementary activities that enhance their overall experience. Customers can review their order history and like the favourite orders to repeat it, and like the feedback on products from others. Additionally the customer can likes any bakery products to save it until next purchase and enhance the other customer shopping process.

System State on Completion:

After adding products to their cart, the customer can review the cart, make adjustments like changing quantities or removing items, and proceed to checkout. Then the system updates the order history and ask to submit feedback.

Scenario 3: Customer gives his Feedback (Ayah Abdalmuti) (MODIFIED)

Initial assumption:
The customer has an account and is logged in, and they have successfully completed a purchase on the online bakery shop.

Normal:
Email Confirmation: After a customer completes a purchase, an automated system sends out an order confirmation email. This email contains a personalised thank-you message, an overview of the order.

Feedback Form Integration: The link directs the customer to a dedicated page on the bakery shop's application, seamlessly integrating the feedback form. The bakery's application ensures a smooth and secure transition from the email to the feedback form.

User-Friendly Feedback System: The feedback form features a straightforward and user-friendly interface with clear fields for customers to rate various aspects of their experience, such as product quality, delivery time, and overall satisfaction.

Comprehensive Input Verification and Error Handling: The system verifies that all necessary fields are filled out before submission and provides clear error messages to assist customers in correcting any issues with their input.

Secure Data Handling: Upon submission, the feedback form securely sends its data to a server, where it is stored in a dedicated database.

Confirmation Message::After the customer successfully submits and the system validates their feedback, a confirmation message is displayed, acknowledging the customer's feedback.

What Can Go Wrong?

Alternative:
If the customer encounters issues during feedback submission, such as unexpected server maintenance, too many people using the system, or server problems, the system automatically notifies both the technical support team and the customer about the problem. Customers receive an apology message for the inconvenience, along with an estimated timeframe for resolution. To ensure smooth operation once the server is stable again, the system sends another email with the feedback link once the server is working again.

Error:
When a customer writes his feedback, he doesn't want to publish his feedback. Therefore, this feedback was recorded and saved in the system without publishing it to the public. But the system behaved wrongly and published his feedback.

Other activities:
1. Feedback Modification Options: Customers have the flexibility to modify or update their feedback using alternative options. The system allows customers to add further comments, provide clarifications, or adjust their ratings.
2. Feedback Medium Choices: Customers are given alternative options for providing feedback beyond traditional forms. The system offers choices such as voice feedback.

System state on completion:
The feedback is securely recorded in the customer feedback database, and the order history is also securely recorded in the system. Additionally, customers receive a confirmation message, fostering a positive relationship with the bakery's online shop.

Scenario 4: Payment processing (Rama Shaheen)

Initial assumptions:

After registering and logging in, customers add products to their shopping carts, customers with loyalty card receive discounts and reward points for each purchase, which improve purchasing experience and encourage regular customers.

Normal:

After making their choice and checking out, customers in the standard payment procedure have a variety of payment options to choose from, including cash on delivery and credit card payments are the choices accessible, when checking out customers shall select their chosen payment option, if the chosen payment method is credit card, the customers are required to enter the card number, expiration date and the security code, following a successful transaction, the system safely processes the payments. Customers must enter their delivery details to complete the order.

Alternative:

There is an extra step in the payment process for customers who have loyalty cards, when a regular customerchooses a payment method, the system recognizes them and recommends using their loyalty card to receive discounts or earn points.

The system will apply the discount to the user's entire purchase if they decide to use their loyalty card, this encourages loyalty card holders to make more purchases on the platform and personalizes the purchasing experience, by providing these advantages, the platform increases customer loyalty, promotes repeat business, and develops a feeling of worth among its users, this extra measure guarantees that loyal customersreceive recognition, which can greatly increase their loyalty and pleasure.

Error:

In the unlikely scenario that problems arise with the online payment process, such as: inaccurate card information or processing delays that exceed 60s, the system will provide them with the choice to either re-enter their payment information accurately or choose another payment method.

Other activities:

If a customer asks for refunds or to return an item, the system responds quickly to the customer request for refunds or returns, the system generates a return product approval number and gives return instructions afterconfirming the purchase data and justifications. The system refunds the original payment and updates inventory records once the items are received. The customer receives an email confirmation detailing the specifics of their refund.

System state on completion:

- After the transaction is finished, the system modifies bank accounts, updates sales and inventory record, plus emails a confirmation including the purchase information, if there is a payment error the order is unconfirmed, no charges are made and the user is informed while the error is noted and customer service is notified. For customers with loyalty cards, the system notifies customers about any changes in their loyalty card.
- Inventory and financial records are updated for returns and refunds, and a confirmation is sent to the customer.

Scenario 5: Admin Edits Product Information (Hala Ziq) (MODIFIED)

Initial assumption:

The admin is logged into the bakery's admin panel, has permission to edit products, and there are products already listed in the system.

Normal:

The admin goes to the "Products" section and uses the search feature to find the specific product they want to edit. The system displays the search results, and the admin selects the desired product. The system then shows the product details, such as name, price, description, and stock levels. The admin makes the necessary changes to the product information and submits them. The system checks the data, saves the changes, and confirms the update.

Alternative:

If the admin decides to cancel the edit, the system will ask for confirmation and discard the changes if confirmed. If the admin selects the wrong product, they can use the search feature again to find and choose the correct one to edit

Error:

If the admin enters invalid data, the system will show an error message, prompting the admin to correct it before saving. If there's a system error, the system will inform the admin that the changes couldn't be saved and may ask them to try again later.

Other activities:

While the admin is making changes in the administration panel, other staff members can view but not edit the inventory.

System state on completion:

If the edit is successful, the product information is updated in the system. If an error occurs, the original product information remains, or the system waits for the issue to be fixed. If the admin cancels the edit, no changes are saved.

2.4. Effort, Time and cost Estimation (Led and Written by Hala, reviewed by Dunia, discussed by all)

- Acronyms: pw= person week; pm= person month; w= week; m= month
- effort= the effort required for a person employed all month/week long (day: 9 hours, week: 5 days, month: 22 days)
- Schedule time = actual time needed to complete the work, based on working days only (without holidays)

NOTE: we assume this project team consist of 3 developers, and the salary for each developer is 700$/month

User Requirement	*Estimated No of Developers* *can work concurrently on UR*	*Estimated Effort* *(average/developers)*	*Total Effort* *(for One Developer)*
UR1.0	2	2 pw	2 * 2 pw = 4 pw
UR2.0	2	3 pw	2 * 3 pw = 6 pw
UR3.0	1	1 pw	1 * 1 pw = 1 pw
UR4.0	1	1 pw	1 * 1 pw = 2 pw
UR5.0	2	4 pw	2 * 4 pw = 8 pw
UR6.0	1	1 pw	1 * 1 pw = 1 pw
UR7.0	2	2 pw	2 *2 pw = 4 pw
UR8.0	1	1 pw	1 * 1 pw = 1 pw
UR9.0	3	3 pw	3 * 3 pw = 9 pw
Avg.Total Effort	15 / 9 = 1.67	18 pw	36 pw

	Min time to complete	Max time to complete
Schedule time 30%	18 * 1.30% = 23.4 w	36 * 1.30 = 46.8 w
cost	Avg.salary/w = 175$	175 * 46.8 = 8190$
Profit Margin *(Min=10%, Max=30%)*	Min cost Max cost	8190 * 1.10 = 9009$ 8190 * 1.30 =10647$

2.5. Actors and Use case (Diagram) Analysis

Actors and Use Cases for the Bakery System (Led and Written by Dunia, discussed by Dunia and Hala, Reviewed by Hala, Modified by Dunia) (MODIFIED).

1. Customer/User:

[1] Register (Verify Phone Number)
[2] Log in
[3] Manage Profile
[4] Browse Product
[5] Search Product
[6] Receive Notification (from SMS for promotions or order updates)
[7] Add Product to Shopping Cart
[8] Submit Feedback
[9] Update Shopping
[10] View Shopping Cart
[11] Confirm Order
[12] View Order Details and History

2. Admin:

[13] View Dashboard
[14] Edit Product Info
[15] View Feedback
[16] Add New Product
[17] Send Notification (via SMS or system)

3. Bank:

[18] Process Payment Transaction (Authorize Payment, Verify Credit Card)
[19] Confirm Successful Payment

4. SMS System:

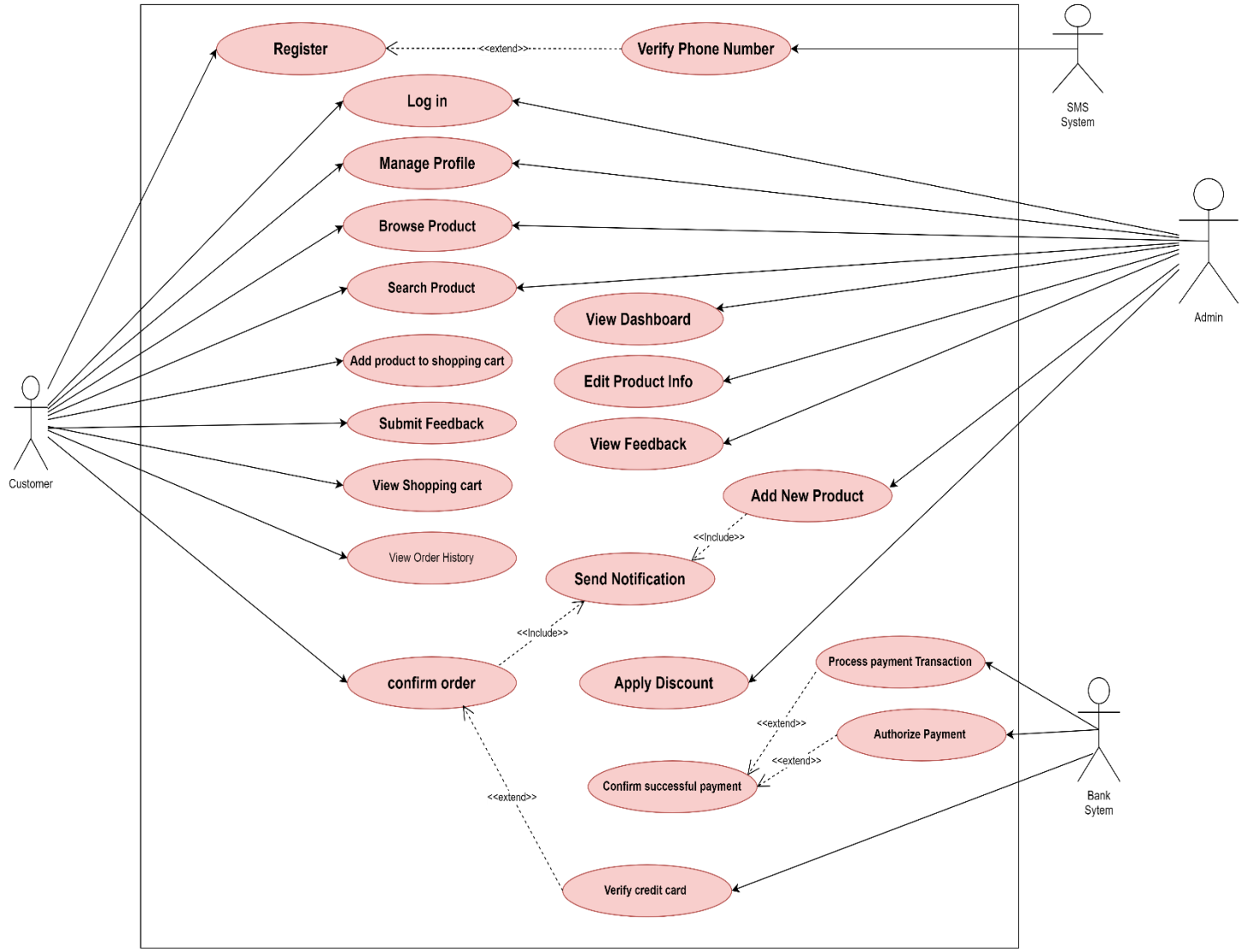
[20] Sends verification messages to verify the phone number.

Actors' roles and their semantic descriptions for the Bakery System (Led and Written by Hala, discussed by Dunia and Hala, Reviewed by Dunia, Modified by Dunia) (MODIFIED).

Actors	Semantics/Description
Customer	The Customer is the primary user of the online bakery shopping online store,they interact with the system to create and manage their accounts, browse andsearch for products, customize orders, add items to their shopping cart, and complete purchases. Customers can also view their order history, provide feedback on products and services, and receive notifications about new products, promotions.
Admin	The admin is responsible for managing the backend operations of the online bakery shopping platform. Admins can monitor and manage products inventory,add and edit product information, and handle restocking. They are also tasked with viewing and responding to customer feedback, also tasked for add new products and promotions, generating business reports, and receiving alerts aboutnew orders and low stock levels to ensure smooth operation of the system.
Bakery system	The Bakery System encompasses the core functionalities required to manage the bakery's online operations. It processes customer orders, ensuring that customization options are correctly applied and reflected in the final order. The system tracks inventory levels in real-time, updating stock statuses to prevent overselling. Generate notification for restocking, receive new order, etc. generating reports to enhance the bakery performance, check and give loyalty points for the regular customers, the Bakery System ensures efficient and accurate handling of orders from placement to delivery.
Bank System	The Bank System handles the financial transactions for the online bakery shopping platform. It processes payments made by customers, confirming or cancelling transactions based on the provided payment details. The Bank System ensures the secure handling of financial information, safe customer datadurring the payment process.
SMS System	The SMS system is integrated into the online bakery shopping platform to sends verification messages during customer registration and login processes to ensuresecure access.

2.6. Use–Case Modelling Diagram: (Led by Hala, Design and discussed by Hala and Dunia, Reviewed by, Ayah and Rama, Ansam, Modified by Hala) (MODIFIED).

UML authoring techniques were applied to visualize and structure the functional requirements of the Bakery System.



2.7. Use-Case Specification

Use Case 1 Specification: sign up and login process (Led and Written by Ansam) (MODIFIED).

Actors	Customer,Admin
Description	The customer uses the system to log in his account or registers for a new one using the system. To use system services and features, this includes providing necessary identification and authentication information.
Pre-condition	1- The customer should be previously registered to log in. 2- The system must be available and operational.
Sequence/flow of event	1- Customer opens the application. 2- Customer selects "Register" or "Log in" from the homepage. 3- Registration Process: • Customer enters name, email, password, confirm password. • System validates: • Customer submits form. • System sends SMS verification code. • Customer enters code. **Alternative**: Provide resend option if SMS fails. **Error**: Prompt for incorrect/expired code. 4- **Login Process:** • Customer enters email, password. • System validates: **Error**: Prompt for incorrect credentials. **5- Verification:** • System verifies entered code. • Success: Registration confirmed. **Error**: Prompt for retry if code is incorrect/expired. **6- Customer logged into application/dashboard.** **Alternative flow :** Users who have already registered on the platform can access the login page by entering their email and password, instead of going through the registration process. Once the system successfully authenticates their credentials, the user is granted access to their account, allowing them to continue browsing and using the platform's services. If the authentication fails, the system will prompt the user to complete the registration process. Additionally, there are alternative ways to interact with the

	platform's functionalities. Users can either search for specific products using the search bar or browse through different categories for a more exploratory shopping experience. For payments, users have the option to pay by cash or by card, providing flexibility in how they complete their transactions. **Error flow:** Possible errors happen when user sends an incorrect or incomplete registration forms. For example, if user leave an empty field or enters incorrectly formatted email address, the system will display the error on screen like (Invalid email format or Password must be at least 8 characters long). The system allow user to request a new SMS and attempt the verification again if the phone number verification fails, for example, because the code that get from SMS was entered incorrectly.
Data	Registration data like Name, Email, Password, Phone number (optional for SMS verification) and Log in data (Email, Password).
Stimulus/Trigger	customer chooses to "Log In" or "Register" on homepage
Post condition/response	1- The customer is registered and logged into their account. 2-The account creation process has successfully stored the new customer information in the system's records. 3-The system's records are updated to reflect the inclusion of the new customer information.
Comments	1-ensure data privacy and security are important process on registration. 2-The user has the option for password recovery and multi-factor authentication to enhance security.

Use Case 2 Specification: Product Browsing (Led and Written by Dunia) (MODIFIED).

Product Browsing	
Actors	Customer
Description	Initiates when a customer logs into the platform intending to browses various bakery products, including viewing detailed product information and customizing selections prior to purchase.
Pre-condition	➢ The customer has successfully logged into thesystem. ➢ The system is operational with a stable network connection and backend.
Sequence/flow of event	**Basic Flow (Normal):** 1. **Access Main Navigation Page**: Customer logs in and accesses the main page where different categories like bread, cakes, and pastries are displayed. 2. **Select Category**: Customer selects a product category to view available products. 3. **View Product Details**: Customer clicks on a desired product, such as a chocolate cake, to view detailed information including price, high-resolution images, description, and ingredients. 4. **Select Customizations**: Customer chooses custom options available for the product, such as size or flavour. 5. **Add to Cart or Continue Browsing**: Customer either adds the product to the cart with selected customizations or continues browsing other categories. **Alternative Flow: Product Out of Stock:** 1. System detects that the selected product or variation is out of stock. 2. System provides two alternatives: a) Customer is redirected to a search functionality to find similar or alternative products using specific criteria or keywords. b) System suggests related products dynamically based on the features of the missing product. **Error Flow:** 1. **System** encounters a failure in retrieving product details due to a server or database malfunction. 2. **System** displays an error message to the **Customer**, informing them of the technical difficulty. 3. **System** suggests trying the action again later. 4. **System** logs the incident for technical analysis and possible rectification. **5.** If the issue persists, **System** temporarily disables the affected product's page to prevent incorrect transactions.
Data	Categories, Product ID, Name, Description, Ingredients, Price, Available Customizations, High-Resolution Images.

Stimulus/Trigger	Logging into the online bakery platform and selecting a product category or interacting with products triggers the system to display detailed information, customization options, and updates based on user actions.
Post-condition/Response	• Order Confirmation: The item(s) are added to the customer's order history. • The system remains stable and ready for further actions, such as proceeding to checkout or continuing to browse other products.
Comments	➢ The system ensures a seamless browsing experience with features like related products and advanced search capabilities, enhancing customer satisfaction by offering easy access to desired products and alternatives.

Use Case 3 Specification: View Order History (Led and Written by Aya) (MODIFIED).

Bakery shop online: Give His Feedback	
Actors	Customer
Description	After the customer opens the website and registered successfully, make an order, after the order process was completed, the system allows customer to give his feedback, the customer able to write good or bad feedback about product quality and the customer is free to publish the feedback or not.
Pre-conditions	A customer has successfully completed a purchase at the bakery shop.
Sequence of events	1. The customer browse for a type of products and confirm the order. 2. After the customer receives the order. 3. The customer opens his profile. 4. Go to the feedback section. 5. Write his feedback about products quality and arrives at order time. 6. Application asked customer to publish his feedback in the public comments section. 7. The customer Choose if he/she needs publish his feedback or not.
Data	Order information (name, quantity, price, ingredients, and images). Feedback information (ratings, comments, date and time).
Trigger	The customer selects the 'Feedback' option from their account menu to provide their feedback."
Post Condition	After receiving the customer's feedback, a thank-you message is sent to acknowledge their input and express gratitude. This action aims to strengthen the relationship between the customer and the bakery shop, enhancing the overall customer experience and encouraging continued interaction.
Comments	The system should ensure that the data is presented securely to protect the privacy of the customer’s transactions.

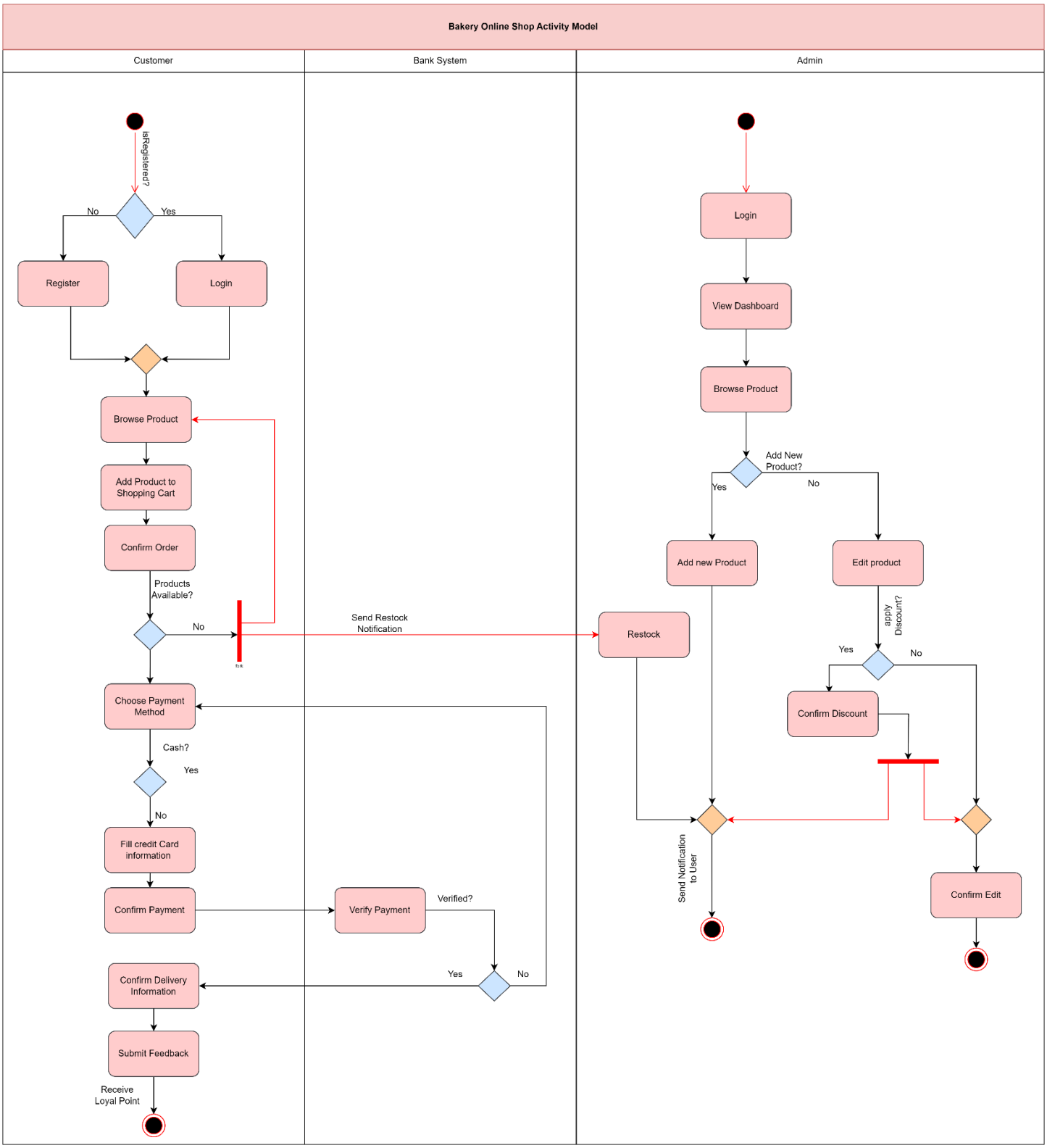
Use Case 4 Specification: Payment Method (Led and Written by Rama).

Scenario 3: payment method	
Actors	Customer
Description	Allows customers to safely finish their purchase transactions and choose a payment option.
Pre-condition	• Client/customer has items in their virtual shopping cart. • Client/customer has given all required shipping details.
Sequence/flow of event	➢ After the customer finishes shopping and selecting their order, the customer hits the "checkout" area. ➢ The customer chooses payment method. (Credit card, loyalty card, cash, etc.) ➢ If credit card, the customer inputs their credit card information and delivery details ➢ If cash, customers shall provide delivery details (address, time of delivery, etc.) ➢ If loyalty card is used along with another payment method, system shall offer discounts and reward points. ➢ The payment is successful and is verified by the system. ➢ The order has been verified by the system, and a confirmation email has been sent to the customer.
Data	○ Payment methods: credit card, cash on delivery, etc. ○ Payment information (card number, expiration date, etc.) ○ Items selected. ○ Delivery details.
Stimulus/Trigger	The customer decides to check out and submit payment.
Post conditions/response	• Payment is processed successfully. • The customer receives loyalty points and discounts for future purchases. • The order is set up for delivery.
Comments	The payment gateway must ensure high security to protect consumer financial information.

Use Case 5 Specification: Admin Edits Product Information (Led and Written by Hala) (MODIFIED).

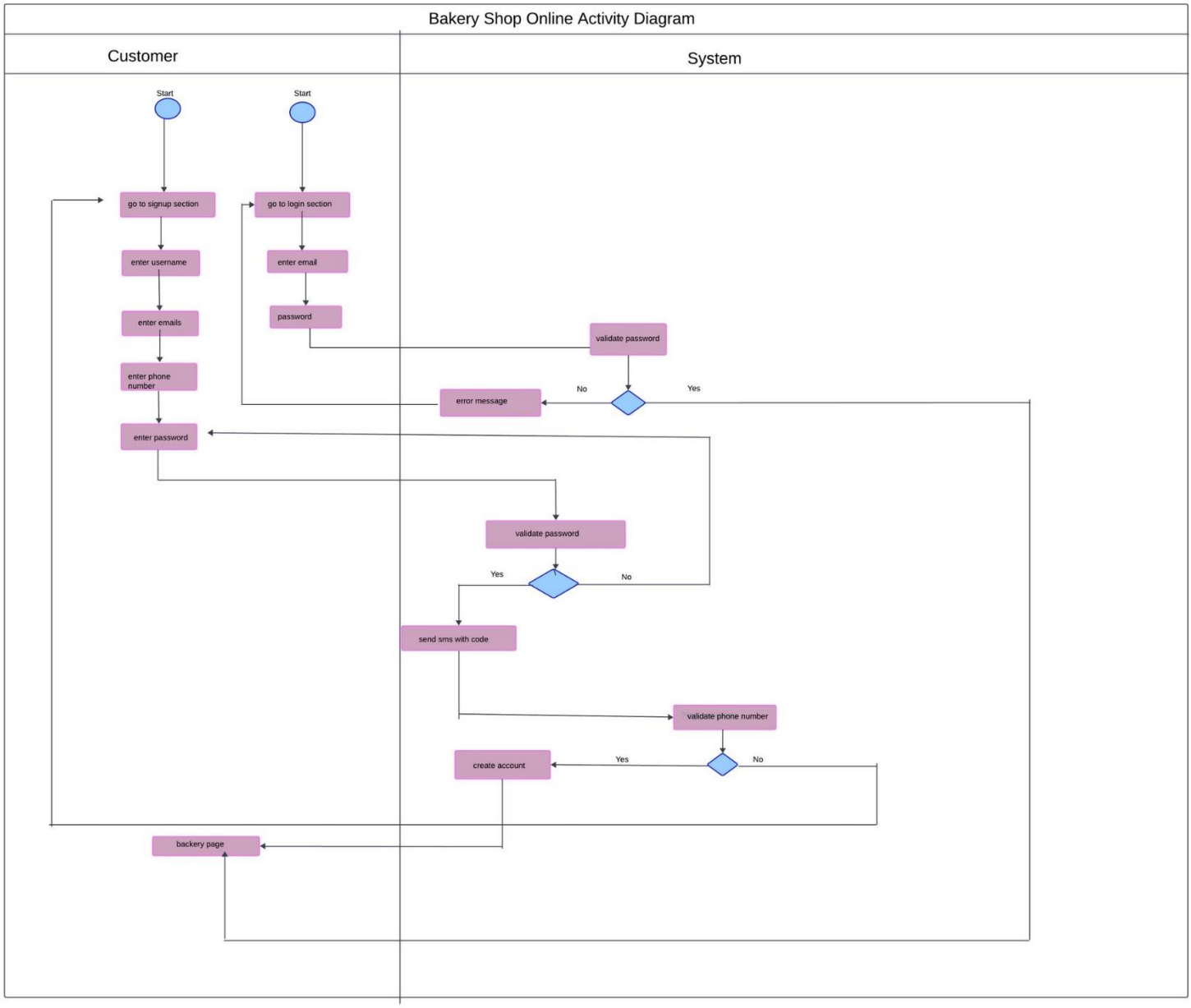
Bakery shop online: Admin Edits Product Information	
Actors	Admin
Description	The admin uses the system to search for and edit product information in the bakery's inventory.
Pre-conditions	➢ The admin is logged into the admin panel. ➢ The admin has the necessary permissions to edit product information. ➢ The system is operational and connected to the product database. ➢ There are existing products in the system to be edited.
Sequence of events	1. Admin navigates to the "Products" section. 2. Admin uses the search feature to find the desired product. 3. System displays search results. 4. Admin selects the product to edit. 5. System shows product details. 6. Admin makes changes to the product information. 7. Admin submits the changes. 8. System validates the data. 9. System saves the changes. 10. System confirms the update. **Alternative Flow:** • If the admin decides to cancel the edit, the system asks for confirmation and discards changes if confirmed. • If the admin selects the wrong product, they can use the search feature again to find and choose the correct one to edit **Error Flow:** • If invalid data is entered, the system shows an error message and prompts the admin to correct it. • If a system error occurs (e.g., database is down), the system informs the admin that changes couldn't be saved and may suggest trying again later or contacting support.
Data	○ Product details (name, price, description, stock levels, etc.) ○ Admin's input data
Trigger	Admin decides to edit the information of an existing product.
Post Condition	- The product information is updated in the system if the edit is successful. - If an error occurs, the original product information remains unchanged. - If the admin cancels the edit, no changes are saved.
Comments	➢ The system logs the admin's actions for audit purposes. ➢ Relevant users may be notified about the product changes. ➢ Error handling for invalid data and system issues is included.

2.8. Activity Modelling (Led and Design by Hala and Dunia, Reviewed by Dunia, Modified by Hala) (MODIFIED).

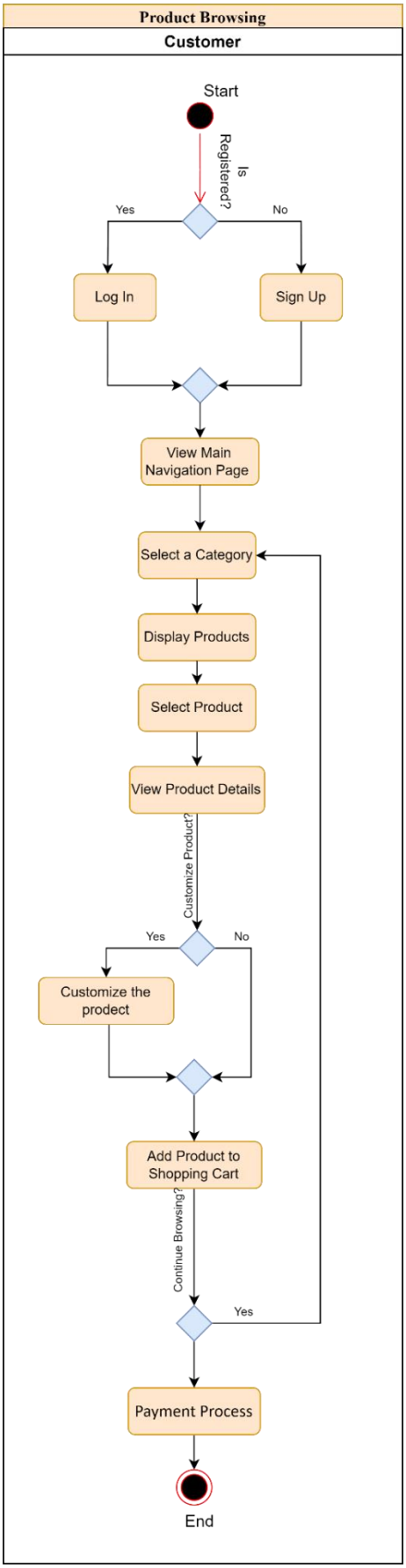


2.9. Use-Case Activity Modelling

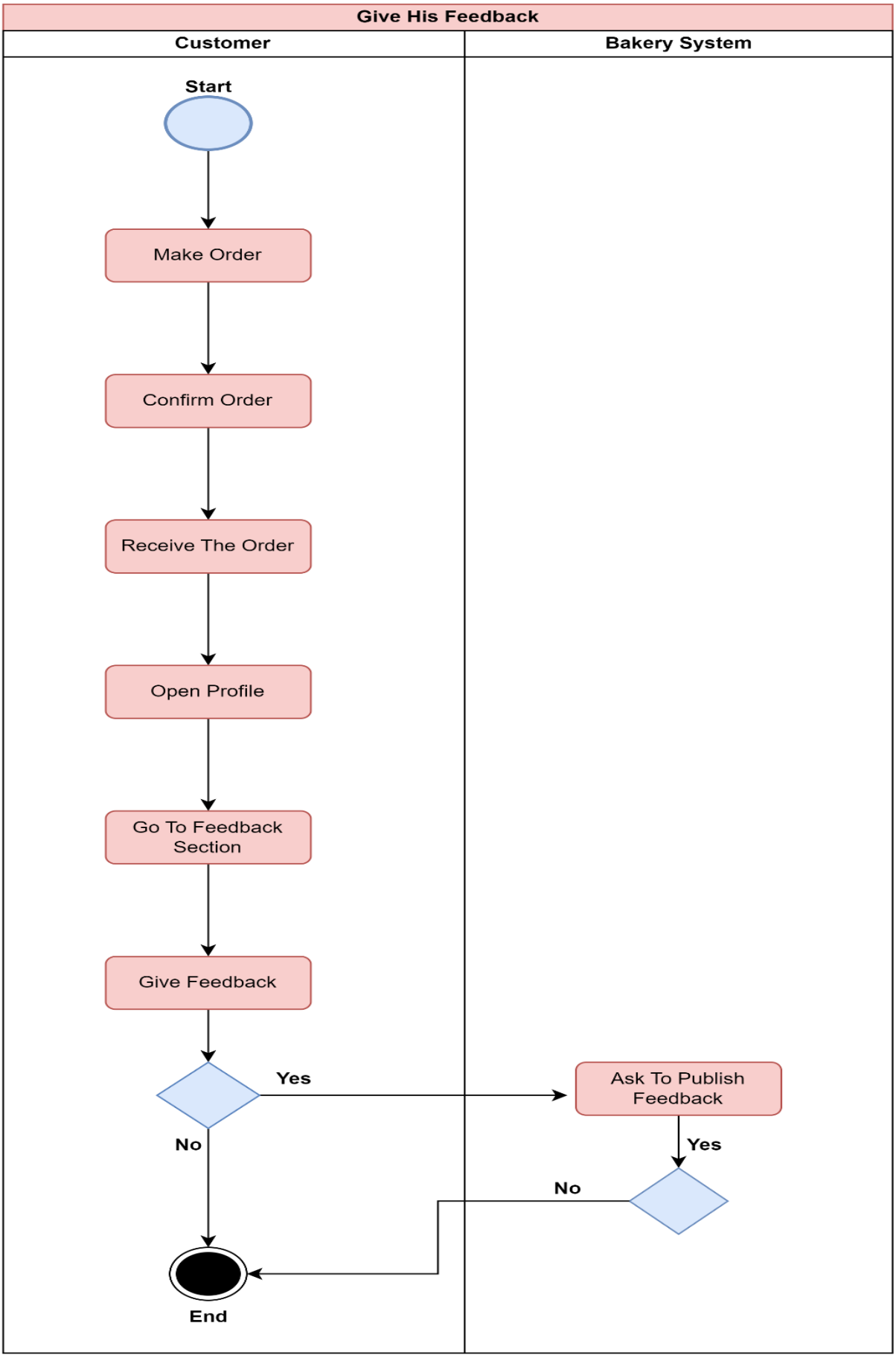
Use Case 1 Activity Modelling: Sign up and Login process (Led and Design by Ansam) (MODIFIED)



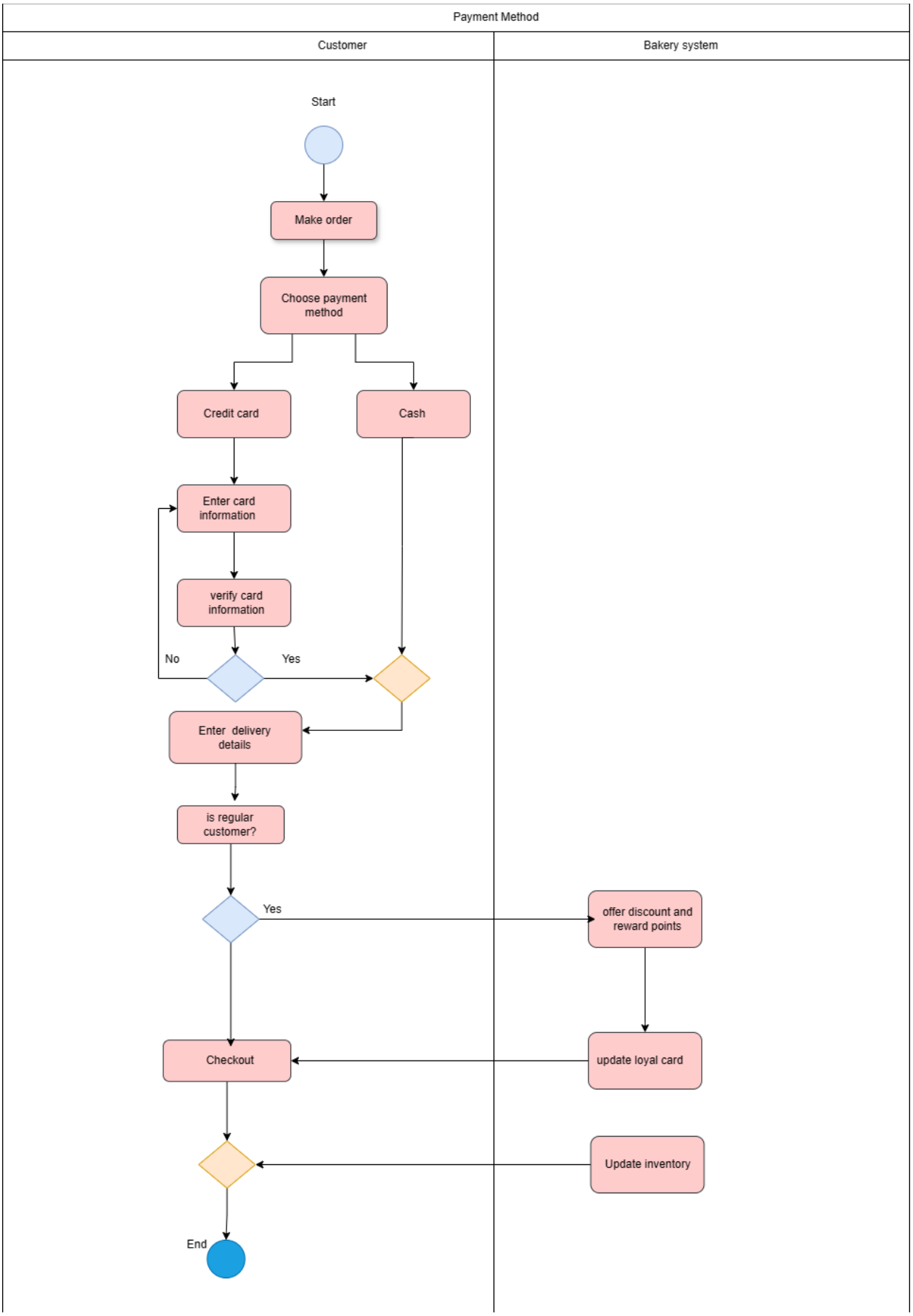
Use Case 2 Activity Modelling: Product Browsing (Led and Design by Dunia) (MODIFIED).



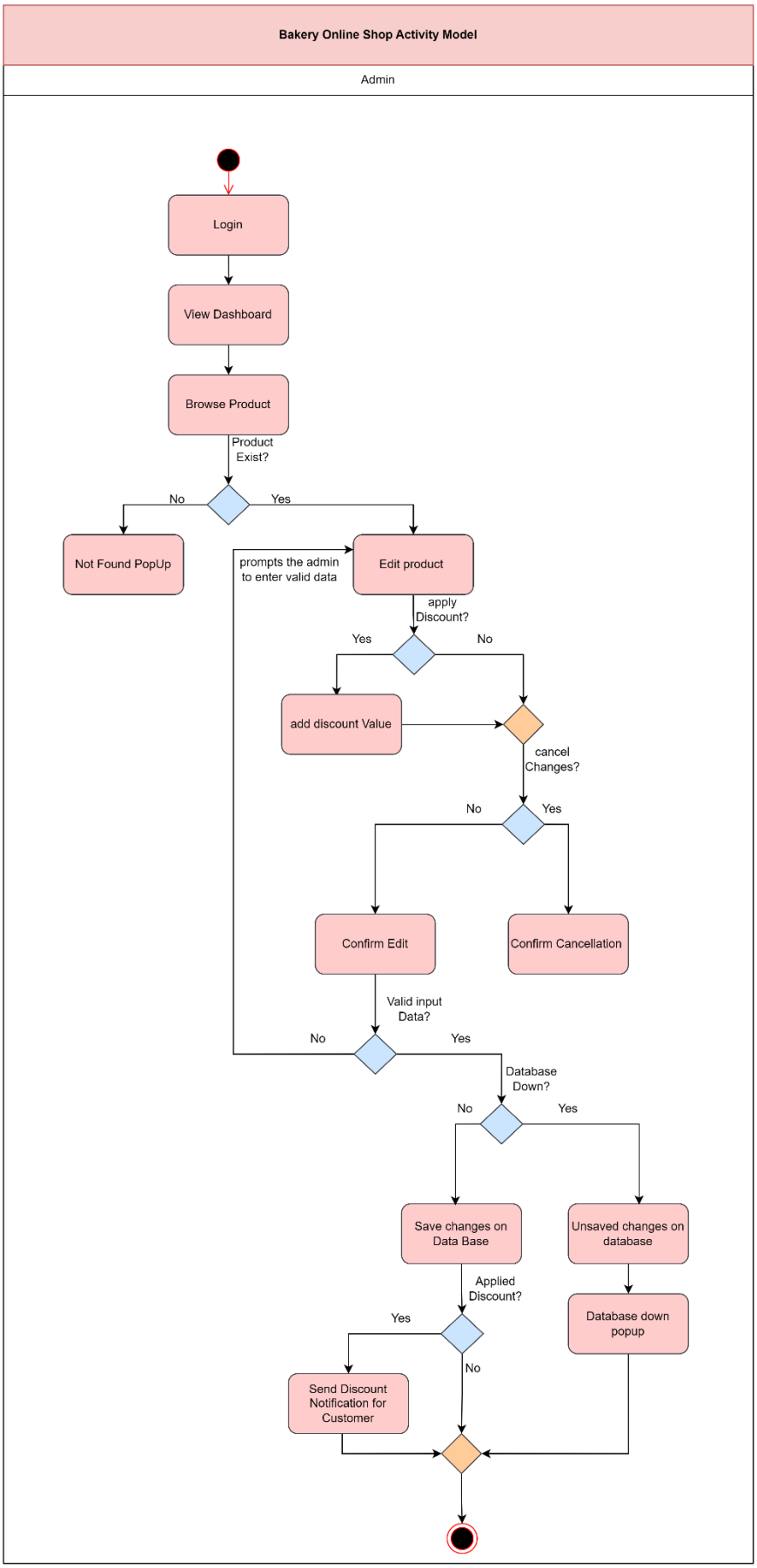
Use Case 3 Activity Modelling: View order history (Led and Design by Ayah) (MODIFIED).



Use Case 4 Activity Modelling: Payment Method (Led and Design by Rama).



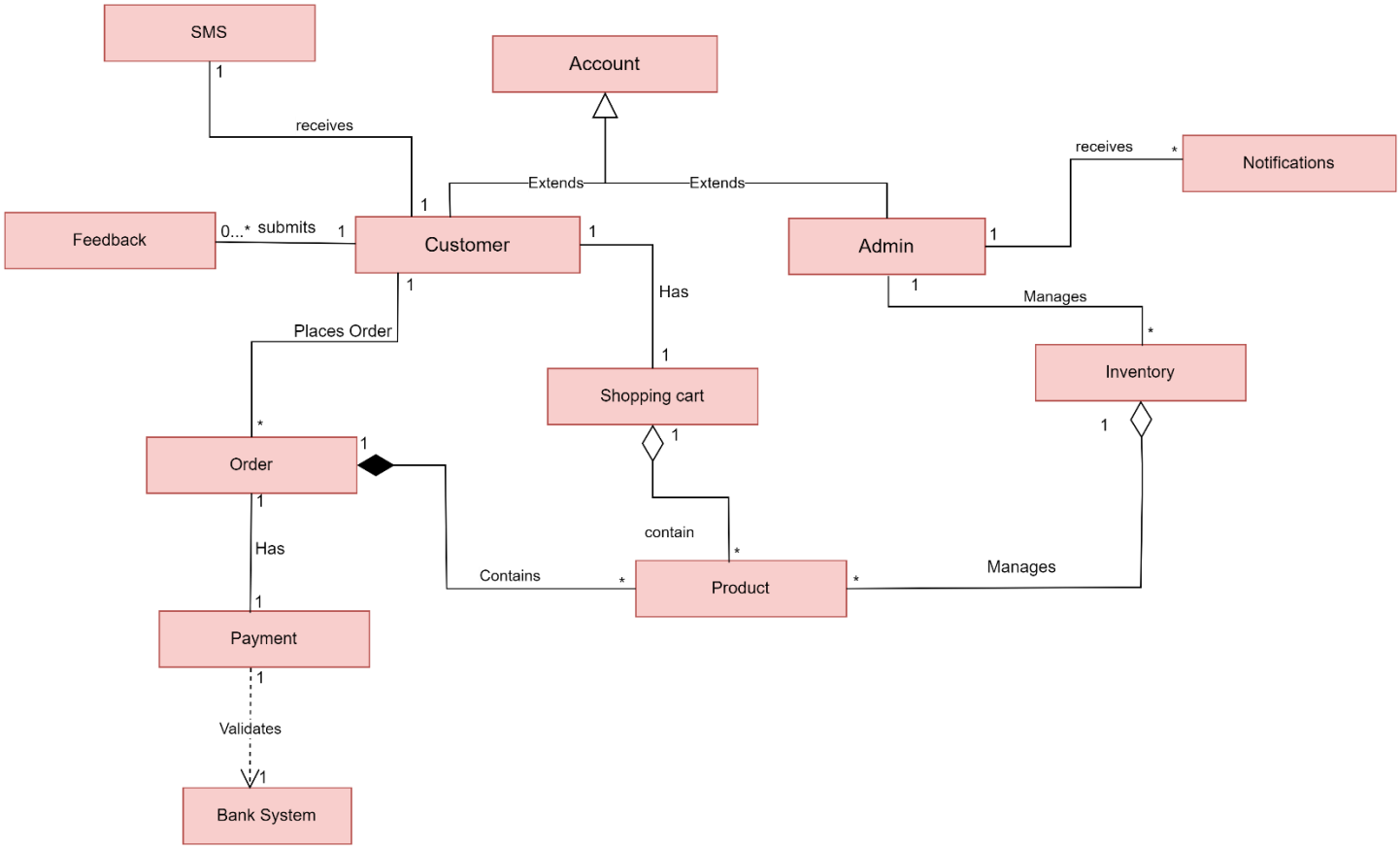
Use Case 5 Activity Modelling: Admin Edits Product Information (Led and Design by Hala) (MODIFIED).



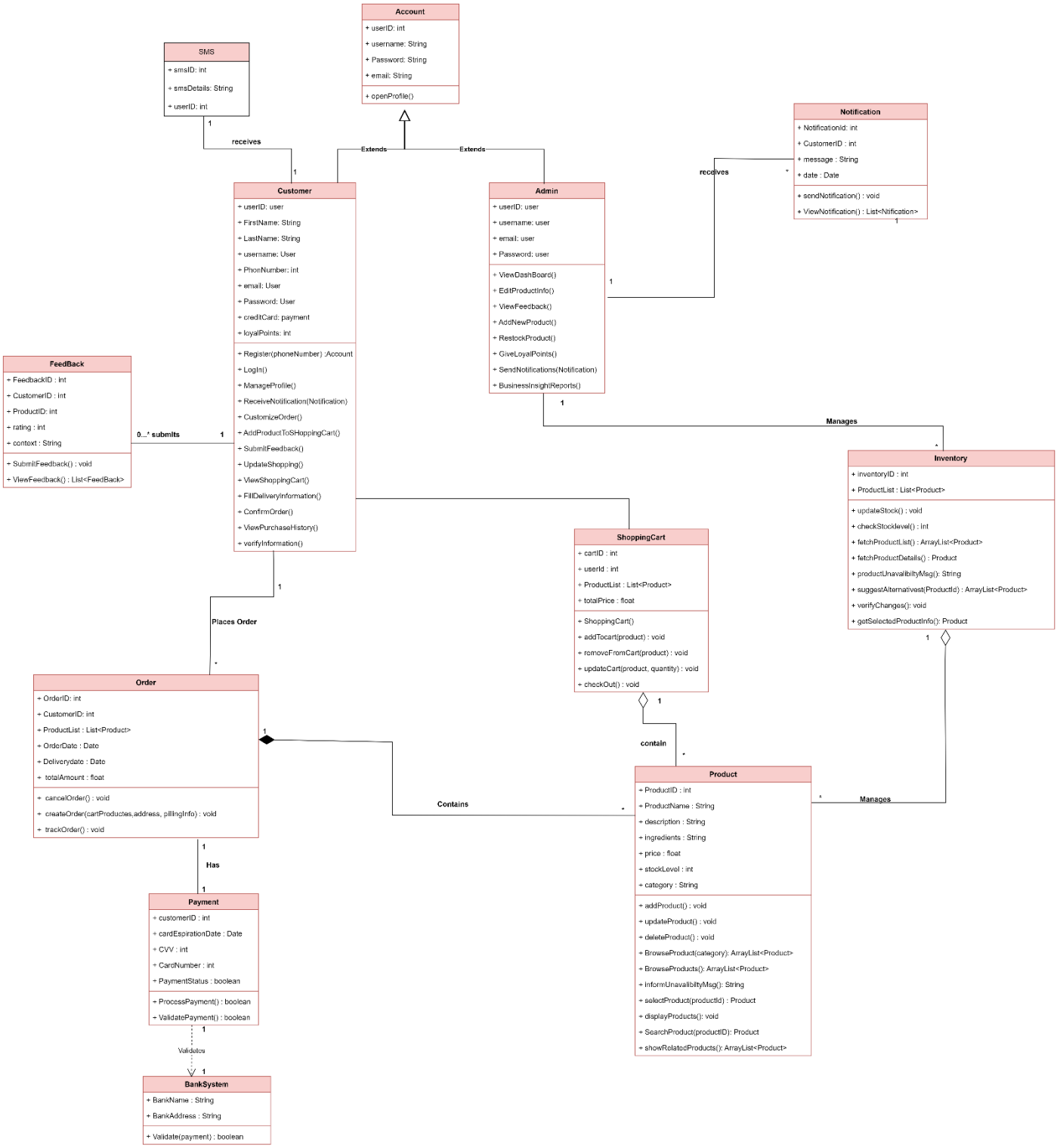
Chapter Three: System Analysis and Modelling.

3.1. System Class modelling and Analysis

Analysis Class Diagram: (Led and Written by Rama and Aya, discussed by Dunia and Hala, Reviewed by Ansam, edited by Dunia and Hala) (MODIFIED).

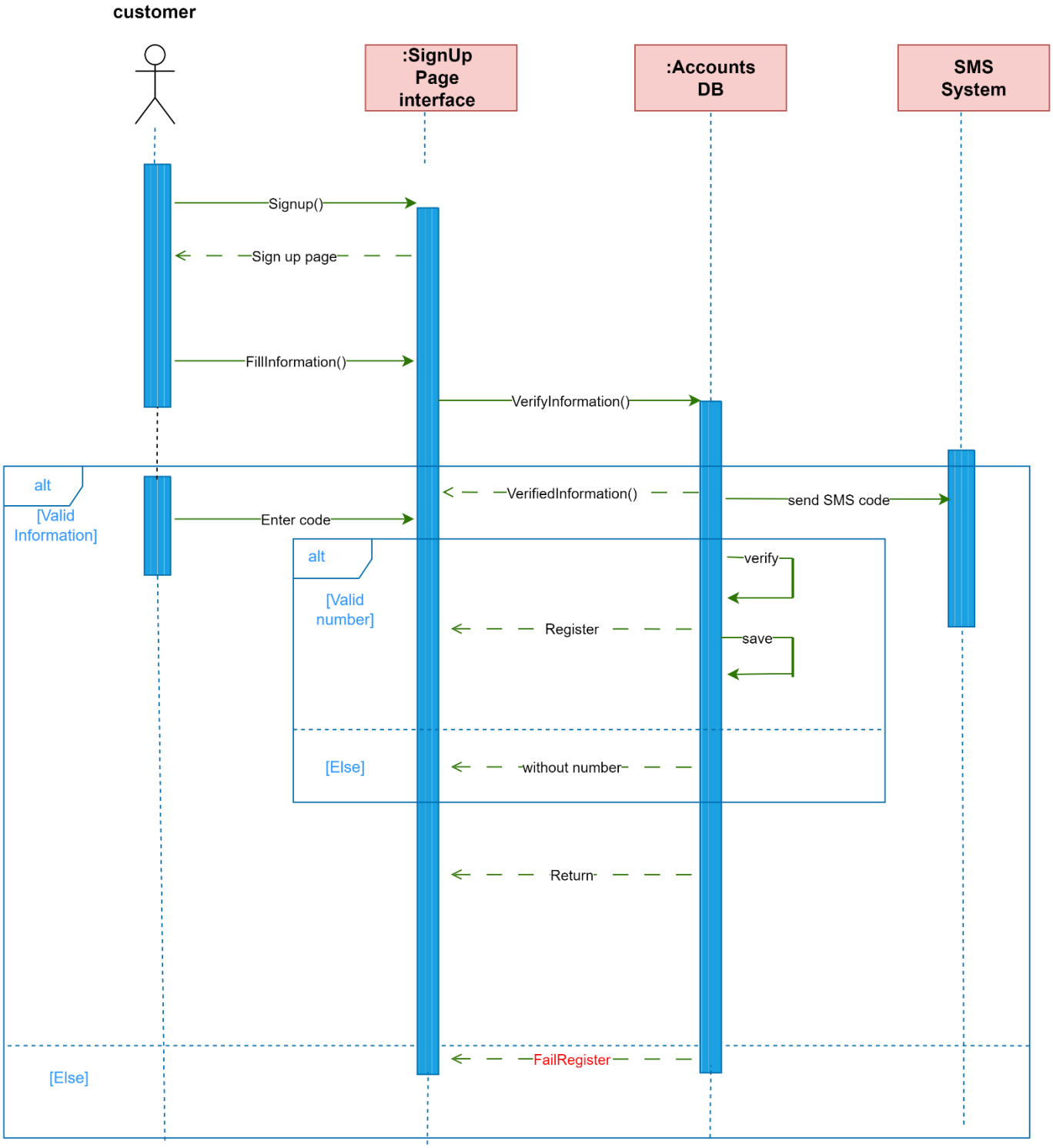


Detailed Class Diagram: (Led and Written by Rama and Aya, discussed by Dunia and Hala, Reviewed by Ansam).

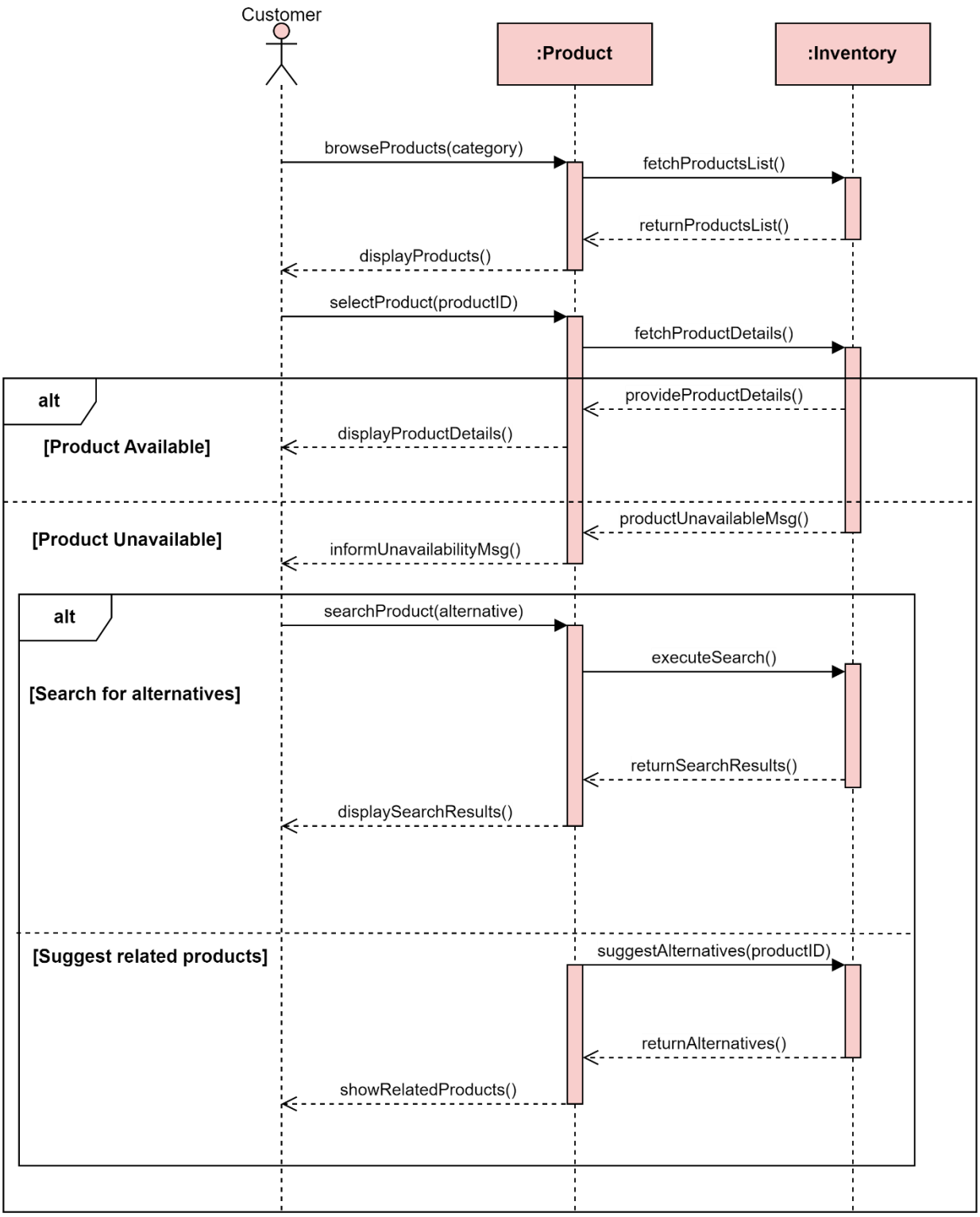


3.2. System Sequence modelling and Analysis

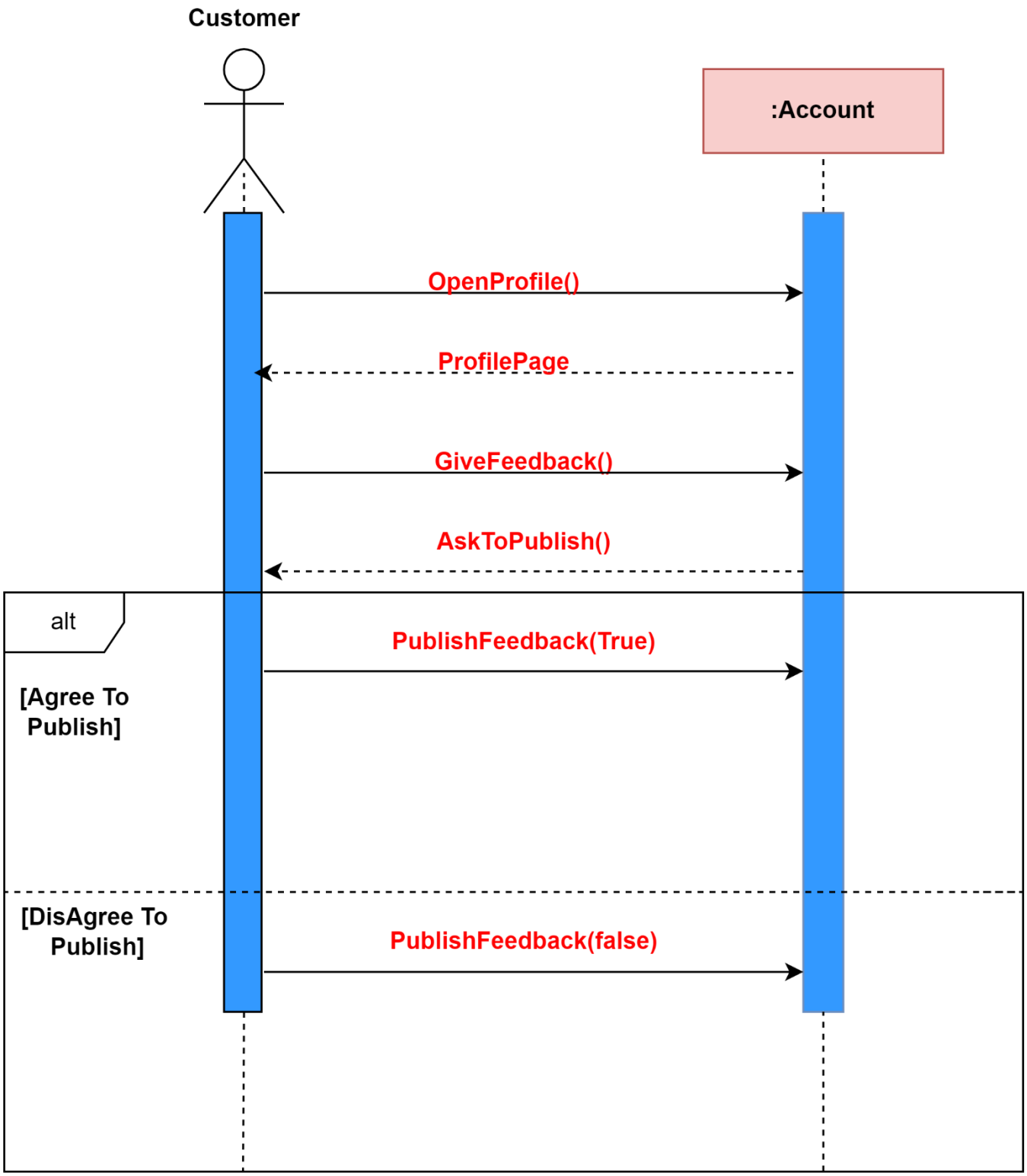
Sequence Diagram 1: Sign Up (Led and Written by Ansam).



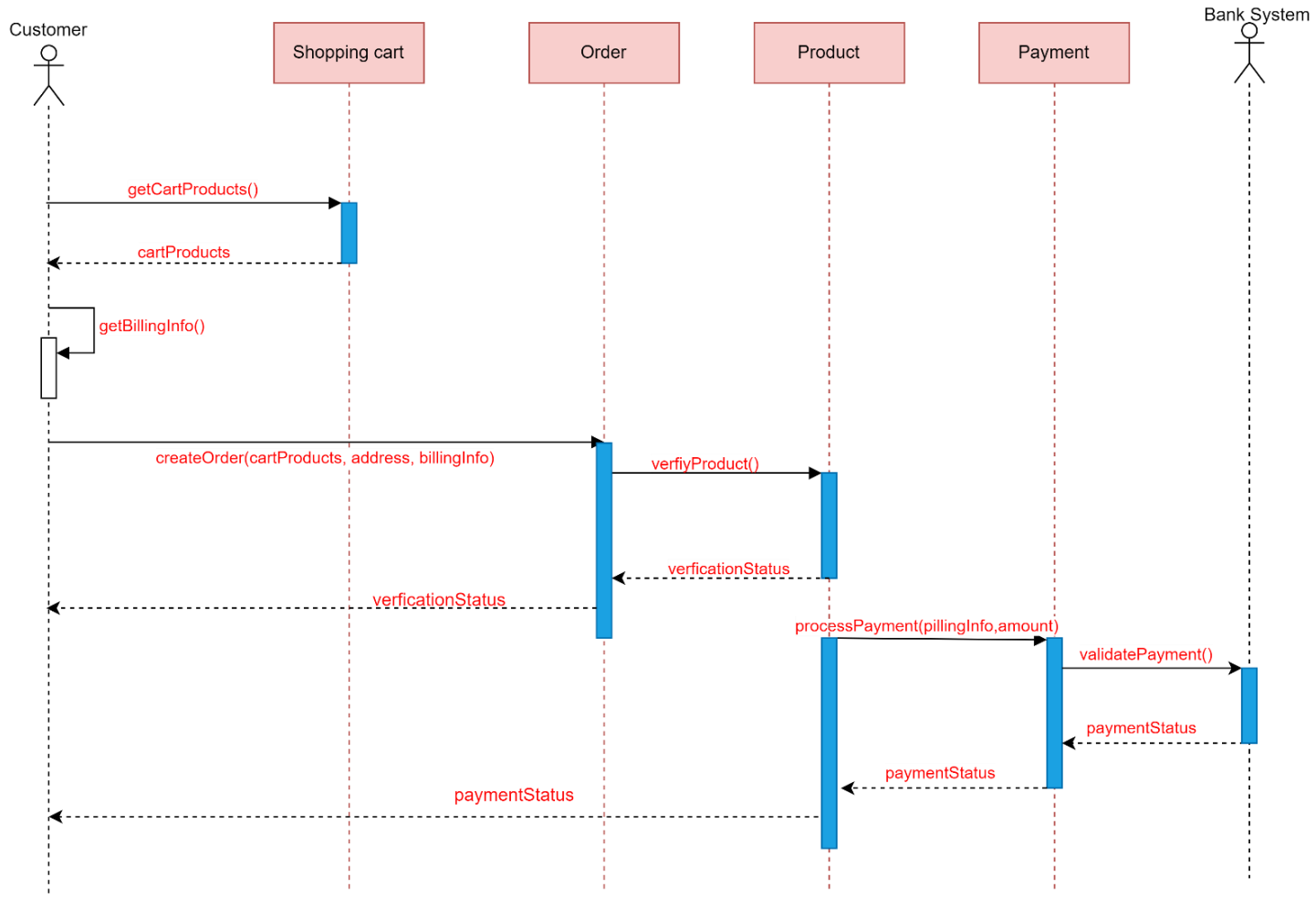
Sequence Diagram 2: Product Browsing (Led and Written by Dunia) (MODIFIED).



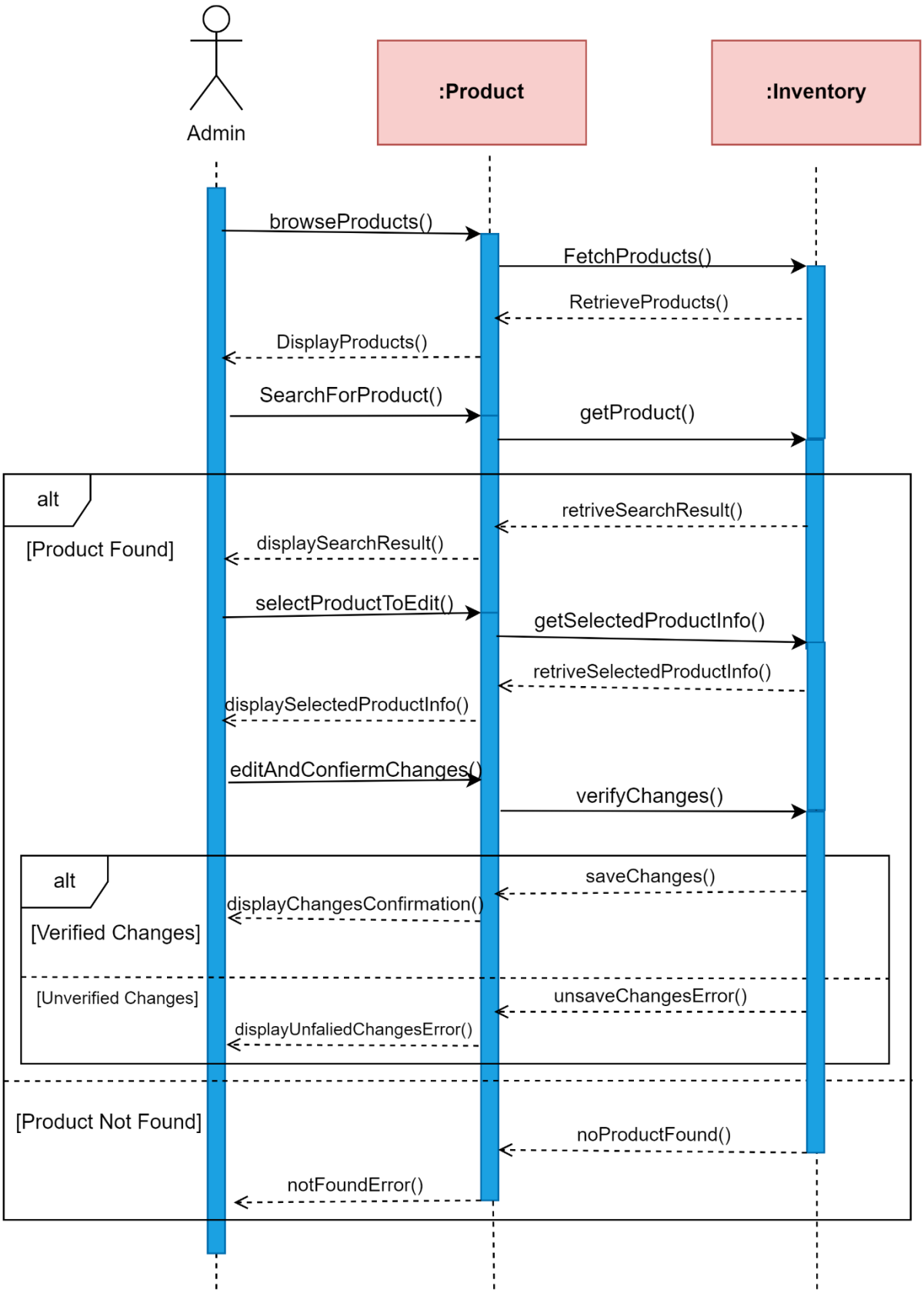
Sequence Diagram 3: View Order History (Led and Written by Aya).



Sequence Diagram 4: Payment processing (Led and Written by Rama).



Sequence Diagram 5: Admin Edits Product Information (Led and Written by Hala).



Chapter Four: System Design and Modelling.

4.1. System Design Goals (Led and Written by Ansam, discussed by all, reviewed by all) (MODIFIED)

General Goals

Low Coupling

We will be using separate models in our system. guaranteeing that any modifications made to one component will have little to no impact on the others .This applied in well-defined interfaces and communication protocols between modules

How to achieve it? : We will use separate models in our system with well-defined interfaces and communication protocols between modules. For example, the Inventory Management and Payment Processing modules will operate independently. This ensures that modifications to the payment methods can be made without affecting other features or the inventory system.

High Cohesion

Every module in our system will be extremely coherent. Meaning that each module will perform a single, well-defined task or a group of related tasks. This design choice makes modules easier to understand, maintain, and test.

How to achieve it? : Each module in our system will perform a single, well-defined task or a group of related tasks. For example, all aspects of user registration, login, profile maintenance, and authentication will be managed by the User Account Maintenance module. This design makes modules easier to understand, maintain, and test.

Specific Goals

Robustness

How to achieve it? : We will use backup databases and regular system health checks to ensure that our system can handle unexpected errors and continue operating smoothly.

Security

Our system handles sensitive data, such as credit card sand financial transactions, it requires a high level of security to ensure data integrity and confidentiality. We will implement robust encryption methods, secure communication protocols to protect user data.

4.2. Component modelling (Led and Written by Hala and Dunia, reviewed by others)

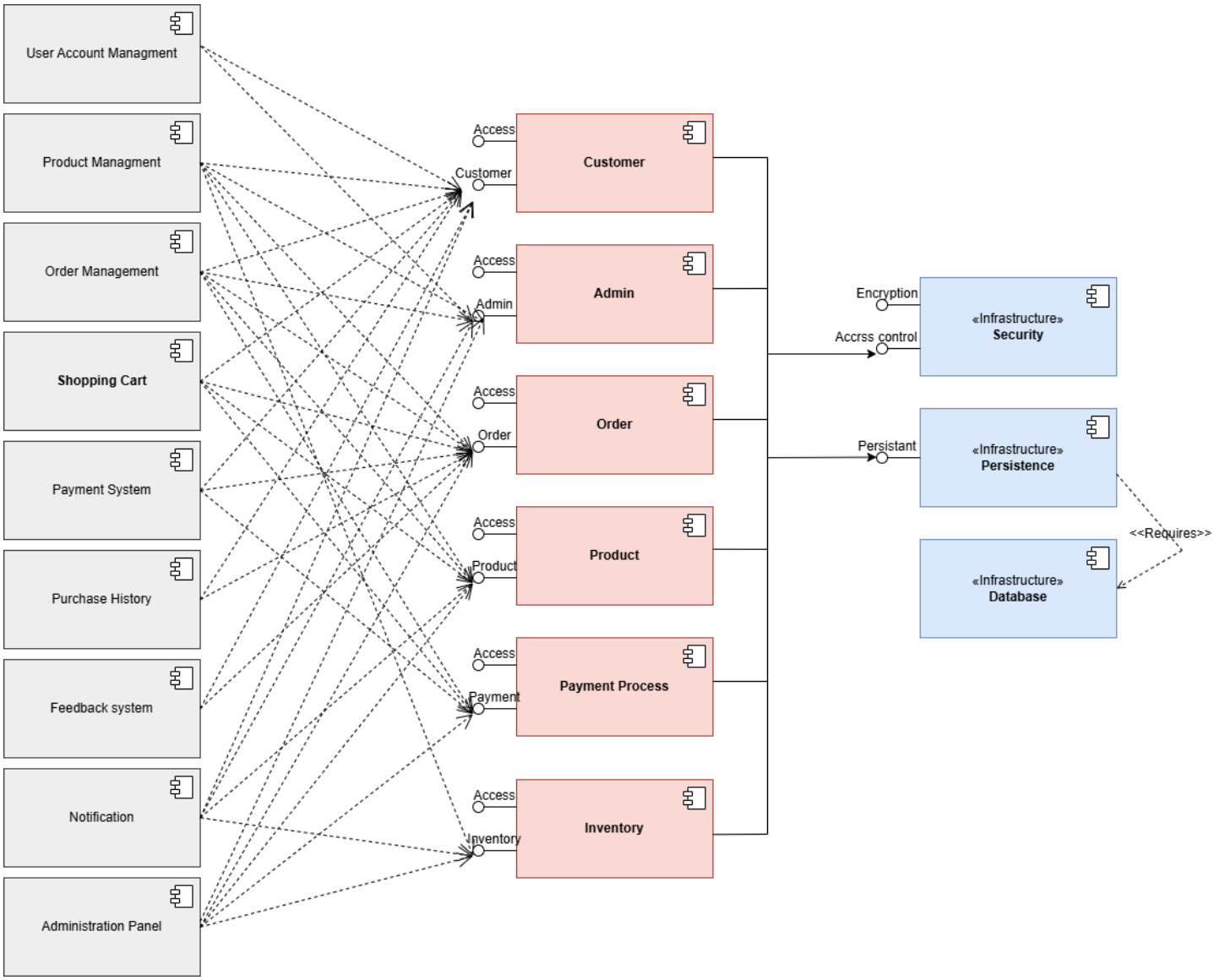
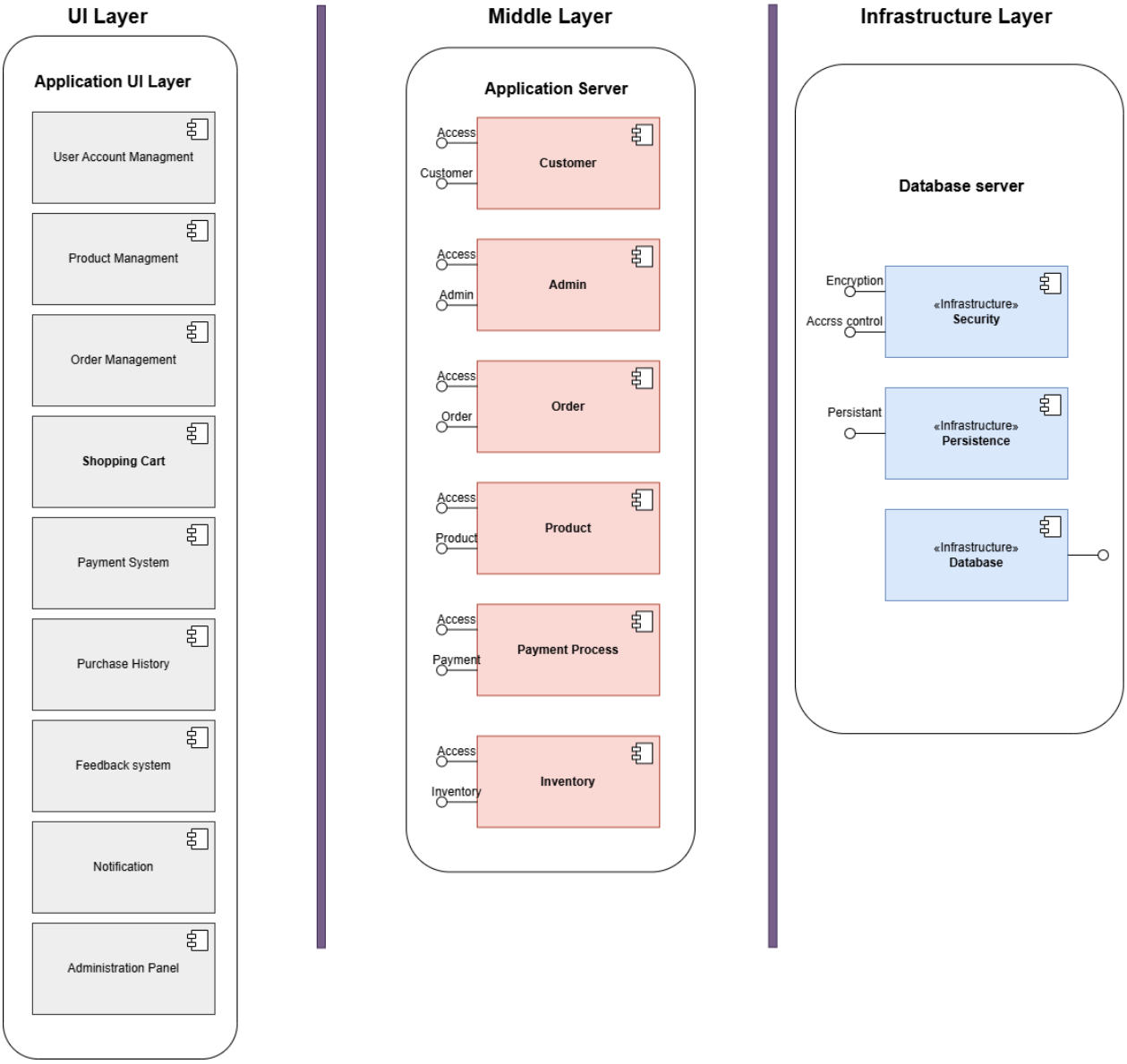


Table 1: Dependencies Table

Component	Customer	Admin	Order	Product	Payment	Inventory
User Account Management	Allows registration, login, and profile management	Manages admin accounts	-	-	-	-
Product Management	Enables browsing, searching, and viewing products	Admins manage product details and availability	Transfers selected products to orders	Manages product details, categories, and stock	-	Provides information on product availability
Order Management	Allows order customization and confirmation	Admins monitor order processing and management	Manages order details	Fetches product details for orders	Initiates payment after order confirmation	-
Shopping Cart	Manages products added to cart by customers	-	Prepares order for checkout	Updates product quantities in the cart	Transfers total price for payment processing	-
Payment System	Processes customer payments	-	Confirms order upon successful payment	-	Manages payment methods (credit card, cash)	-
Purchase History	Allows customers to view past orders	-	Displays order history	-	-	-
Feedback and Rating System	Customers submit feedback on orders and products	-	Links feedback to specific orders	-	-	-
Notification System	Sends updates to customers about orders and products	Sends alerts to admins about orders, stock levels	-	Alerts customers about new products and discounts	-	Alerts admins about low stock levels
Administration Panel	-	Admins manage all aspects of the system	Admins monitor order processing	Admins manage product inventory	Admins review payment reports	Admins manage inventory, restocking, and stock

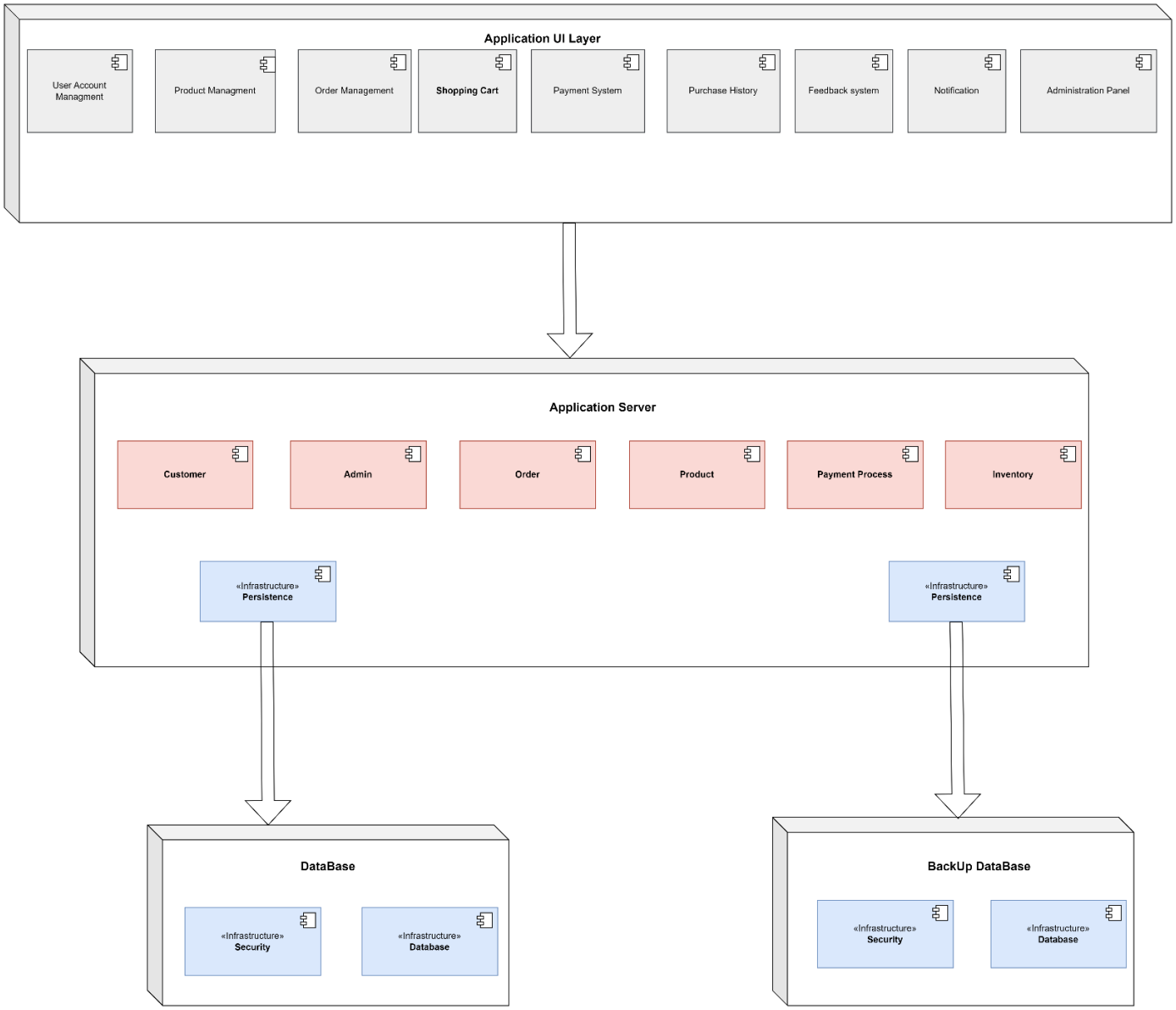
4.3. System and architectural Design (Led and Written by Hala and Dunia, reviewed by others)



We chose a layered architecture for our baker shop online system due to its effectiveness in meeting our design goals and system non-functional requirements.

- **Layered architecture inherently supports low coupling** by separating the system into distinct layers (UI, Application, Infrastructure). Each layer operates independently, and communication between layers occurs through well-defined interfaces. For example, the Payment Processing module in the Application Layer can be modified or upgraded without affecting the Inventory Management module, as their interactions are mediated through clearly defined APIs or service interfaces.
- **Each layer and module perform a specific set of related tasks, ensuring high cohesion.** For example, the User Account Maintenance module is responsible only for user-related functionalities (registration, login, profile management, authentication), ensuring that all related tasks are handled in one cohesive unit.
- **Regular health checks and backup databases** are managed at the infrastructure level, ensuring the system can handle unexpected errors and continue operating smoothly. For example, the use of redundant databases and regular system monitoring within the Infrastructure Layer guarantees continuous operation even during unexpected issues.
- **Robust security measures can be implemented at the infrastructure level,** ensuring secure data handling across all layers. For example, Encryption and secure communication protocols are managed within the Infrastructure Layer, protecting sensitive data such as credit card information and financial transactions across the system.

4.4. System and Deployment modelling (Led and Written by Ansam, reviewed by others)



Note: All Diagrams designed by using an online UML tool: [Drow.io]